Year 8

# REVISION PACK

## End of Unit 4 Test

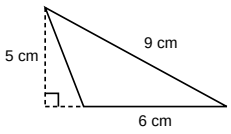
**Name:** _____

**Class:** _____

# Unit 4 · Area of a triangle

## Worked example: the area of a triangle

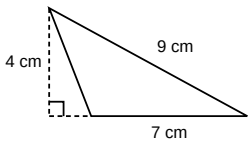
Find the area of this triangle.



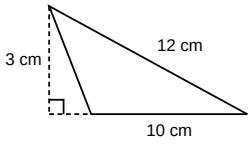

| | | |
|---|---|---|
| **Step 1** | Use base and height (at right angles) | base $6$, height $5$ |
| **Step 2** | Half × base × height | $\frac{1}{2} \times 6 \times 5$ |
| **Step 3** | Answer | $15\ \text{cm}^2$ |

## Practice: the area of a triangle

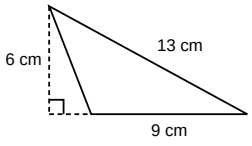
**1.1** Find the area of this triangle.



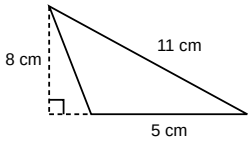

**1.2** Find the area of this triangle.



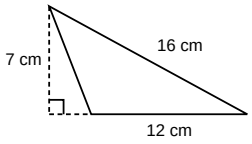

**1.3** Find the area of this triangle.



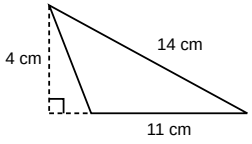

**1.4** Find the area of this triangle.




**1.5** Find the area of this triangle.



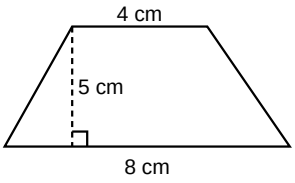

**1.6** Find the area of this triangle.



## Unit 4 · Area of a trapezium

### Worked example: the area of a trapezium

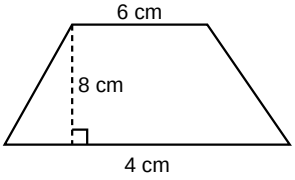
Find the area of the trapezium.




| | | |
|---|---|---|
| **Step 1** | Add the parallel sides | $4 + 8 = 12$ |
| **Step 2** | Halve | $12 \div 2 = 6$ |
| **Step 3** | Times the height | $6 \times 5 = 30$ |
| **Step 4** | Answer | $30\ \text{cm}^2$ |

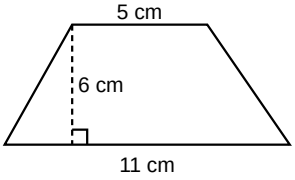
### Practice: the area of a trapezium

**2.1**



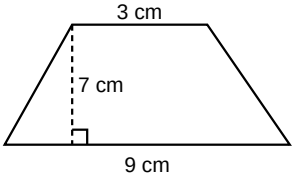

Find the area of the trapezium.

**2.2**



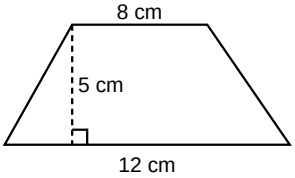

Find the area of the trapezium.

**2.3**



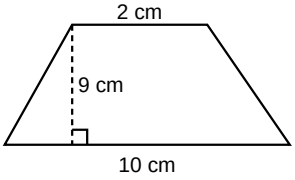

Find the area of the trapezium.

**2.4**



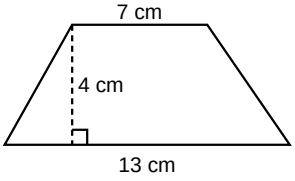

Find the area of the trapezium.

**2.5**




Find the area of the trapezium.

**2.6**




Find the area of the trapezium.

## Unit 4 · Volume and surface area

**Worked example: volume and surface area of a cuboid**

A cuboid has edges of length 6 cm, 4 cm and 2 cm.

(a) Work out the volume. (b) Work out the total surface area.

| | | | |
|---|---|---|---|
| (a) | **Step 1** | Length × width × height | $6 \times 4 \times 2 = 48$ |
| | **Step 2** | Answer | $48\ \text{cm}^3$ |
| (b) | **Step 1** | Area of the 3 different faces | $6 \times 4 = 24,\ 6 \times 2 = 12,\ 4 \times 2 = 8$ |
| | **Step 2** | Each face twice | $2 \times (24 + 12 + 8) = 88$ |
| | **Step 3** | Answer | $88\ \text{cm}^2$ |

### Practice: volume and surface area of a cuboid

**3.1** A cuboid has edges of length 5 cm, 6 cm and 4 cm.

(a) Work out the volume. (b) Work out the total surface area.

**3.2** A cuboid has edges of length 9 cm, 3 cm and 2 cm.

(a) Work out the volume. (b) Work out the total surface area.

**3.3** A cuboid has edges of length 7 cm, 7 cm and 5 cm.

(a) Work out the volume. (b) Work out the total surface area.

**3.4** A cuboid has edges of length 10 cm, 6 cm and 3 cm.

(a) Work out the volume. (b) Work out the total surface area.

**3.5** A cuboid has edges of length 8 cm, 5 cm and 4 cm.

(a) Work out the volume. (b) Work out the total surface area.

**3.6** A cuboid has edges of length 11 cm, 4 cm and 2 cm.

(a) Work out the volume. (b) Work out the total surface area.

### Worked example: litres and units of volume

A fish tank is a cuboid. The base measures 40 cm by 25 cm. Water is poured in to a depth of 10 cm.

(a) Work out the amount of water in litres. (b) Change 7 cm³ to mm³. (c) Show that 2.3 m³ = 2300 litres.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Volume | $40 \times 25 \times 10 = 10\,000\ \text{cm}^3$ |
| | Step 2 | 1 litre = 1000 cm³ | $10\,000 \div 1000 = 10$ |
| | Step 3 | Answer | $10\ \text{litres}$ |
| (b) | Step 1 | 1 cm³ = 1000 mm³ | $7 \times 1000 = 7000\ \text{mm}^3$ |
| (c) | Step 1 | 1 m³ = 1000 litres | $2.3 \times 1000 = 2300\ \text{litres}$ |

## Practice: litres and units of volume

**4.1** A fish tank is a cuboid. The base measures 45 cm by 30 cm. Water is poured in to a depth of 20 cm.

(a) Work out the amount of water in litres. (b) Change 50 cm³ to mm³. (c) Show that 4.6 m³ = 4 600 litres.

**4.2** A fish tank is a cuboid. The base measures 60 cm by 30 cm. Water is poured in to a depth of 25 cm.

(a) Work out the amount of water in litres. (b) Change 8 cm³ to mm³. (c) Show that 3.2 m³ = 3 200 litres.

**4.3** A fish tank is a cuboid. The base measures 50 cm by 40 cm. Water is poured in to a depth of 15 cm.

(a) Work out the amount of water in litres. (b) Change 12 cm³ to mm³. (c) Show that 1.75 m³ = 1 750 litres.

**4.4** A fish tank is a cuboid. The base measures 80 cm by 25 cm. Water is poured in to a depth of 30 cm.

(a) Work out the amount of water in litres. (b) Change 4.5 cm³ to mm³. (c) Show that 6.8 m³ = 6 800 litres.

**4.5** A fish tank is a cuboid. The base measures 35 cm by 20 cm. Water is poured in to a depth of 40 cm.

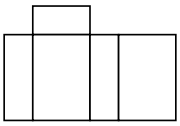
(a) Work out the amount of water in litres. (b) Change 30 cm³ to mm³. (c) Show that 0.9 m³ = 900 litres.

**4.6** A fish tank is a cuboid. The base measures 70 cm by 30 cm. Water is poured in to a depth of 20 cm.

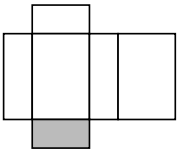
(a) Work out the amount of water in litres. (b) Change 2.6 cm³ to mm³. (c) Show that 4.25 m³ = 4 250 litres.

## Unit 4 · Nets

### Worked example: completing a net and naming the solid

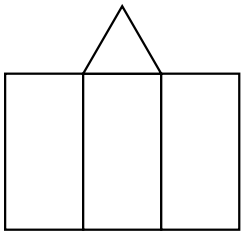


The diagram shows an incomplete net of a solid shape.
(a) Copy the net and add a face to complete it.
(b) What is the name of this solid?
(c) Sketch the solid.

| | | | |
|---|---|---|---|
| (a) | **Step 1** Count the faces | 5 rectangles, 1 missing |  |
| | **Step 2** Answer: add the shaded face | | |
| (b) | **Step 1** 6 faces, all rectangles | cuboid | |
| (c) | **Step 1** Draw a box shape | show the hidden edges dashed | |

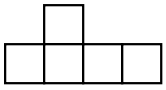
## Practice: completing a net and naming the solid

**5.1**



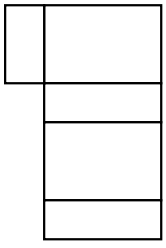
The diagram shows an incomplete net of a solid shape.
(a) Copy the net and add a face to complete it.
(b) What is the name of this solid?
(c) Sketch the solid.

**5.2**



The diagram shows an incomplete net of a solid shape.
(a) Copy the net and add a face to complete it.
(b) What is the name of this solid?
(c) Sketch the solid.

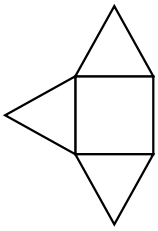
**5.3**



The diagram shows an incomplete net of a solid shape.
(a) Copy the net and add a face to complete it.
(b) What is the name of this solid?
(c) Sketch the solid.

## Practice: completing a net and naming the solid (continued)

**5.4**



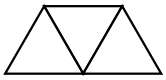
The diagram shows an incomplete net of a solid shape.

(a) Copy the net and add a face to complete it.

(b) What is the name of this solid?

(c) Sketch the solid.

**5.5**



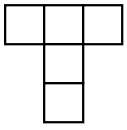
The diagram shows an incomplete net of a solid shape.

(a) Copy the net and add a face to complete it.

(b) What is the name of this solid?

(c) Sketch the solid.

**5.6**



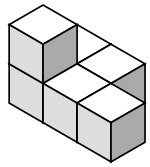
The diagram shows an incomplete net of a solid shape.

(a) Copy the net and add a face to complete it.

(b) What is the name of this solid?

(c) Sketch the solid.

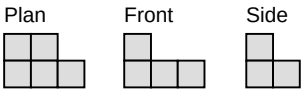
### Worked example: drawing plans and elevations



front side

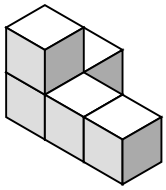
This model is made from 1 cm cubes. Draw the plan, the front elevation and the side elevation.

| | | |
|---|---|---|
| **Step 1** | Plan: look down from above | squares where there are cubes |
| **Step 2** | Front: look from the front | column heights $2, 1, 1$ |
| **Step 3** | Side: look from the side | column heights $2, 1$ |
| **Step 4** | Answer | |




## Practice: drawing plans and elevations

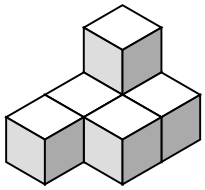
**6.1**



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

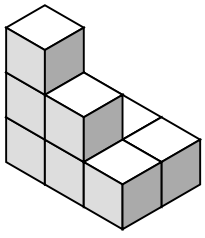
**6.2**



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

**6.3**

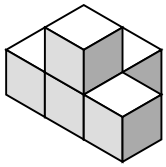


front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

## Practice: drawing plans and elevations (continued)

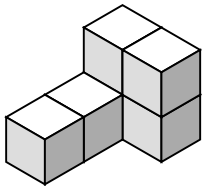
**6.4**



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

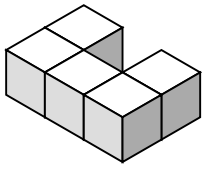
**6.5**



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

**6.6**



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

### Worked example: a block with a hole

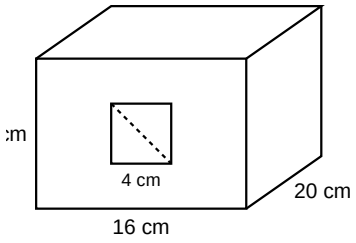
A block of wood is 10 cm wide, 8 cm high and 15 cm long. A square hole, with sides 2 cm, goes all the way through it.

(a) Work out the volume of the wood.
(b) Work out the total surface area of the wood.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Whole block | $10 \times 8 \times 15 = 1200$ |
| | Step 2 | The hole | $2 \times 2 \times 15 = 60$ |
| | Step 3 | Answer | $1200 - 60 = 1140\ \text{cm}^3$ |
| (b) | Step 1 | Front and back | $2 \times (80 - 4) = 152$ |
| | Step 2 | Top and bottom | $2 \times 10 \times 15 = 300$ |
| | Step 3 | Two sides | $2 \times 8 \times 15 = 240$ |
| | Step 4 | Inside the hole | $4 \times 2 \times 15 = 120$ |
| | Step 5 | Answer | $152 + 300 + 240 + 120 = 812\ \text{cm}^2$ |

## Practice: a block with a hole

**7.1**

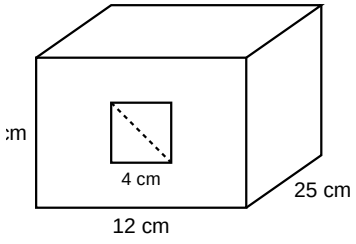



A block of wood has a square hole, with sides 4 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

**7.2**

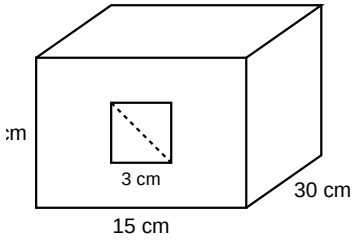



A block of wood has a square hole, with sides 4 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

**7.3**



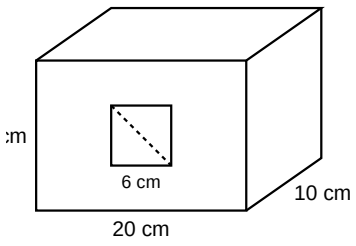

A block of wood has a square hole, with sides 3 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

## Practice: a block with a hole (continued)

**7.4**

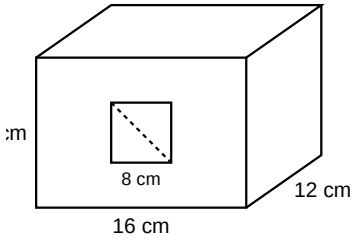



A block of wood has a square hole, with sides 6 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

**7.5**

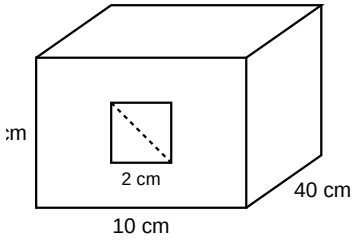



A block of wood has a square hole, with sides 8 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

**7.6**




A block of wood has a square hole, with sides 2 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

**Worked example: the area of a path**

A square lawn has sides of 8 m. A path 50 cm wide goes all the way round it. Work out the area of the path in m².

| | | |
|---|---|---|
| **Step 1** | Change cm to m | $50\text{ cm} = 0.5\text{ m}$ |
| **Step 2** | Side of lawn and path | $8 + 0.5 + 0.5 = 9$ |
| **Step 3** | Big square | $9 \times 9 = 81$ |
| **Step 4** | Lawn | $8 \times 8 = 64$ |
| **Step 5** | Subtract | $81 - 64 = 17$ |
| **Step 6** | Answer | $17\text{ m}^2$ |

## Practice: the area of a path

**8.1** A square lawn with sides of 9 m has a path 50 cm wide all the way round it. Work out the area of the path in m².

**8.2** A square lawn with sides of 7 m has a path 100 cm wide all the way round it. Work out the area of the path in m².

**8.3** A rectangular lawn 10 m by 6 m has a path 50 cm wide all the way round it. Work out the area of the path in m².

**8.4** A square lawn with sides of 15 m has a path 100 cm wide all the way round it. Work out the area of the path in m².

**8.5** A rectangular lawn 5 m by 12 m has a path 50 cm wide all the way round it. Work out the area of the path in m².

**8.6** A square lawn with sides of 3 m has a path 25 cm wide all the way round it. Work out the area of the path in m².

**Worked example: a parallelogram and a triangle with the same area**

A parallelogram has base 9 cm and height 4 cm. A triangle has the same area and a height of 6 cm. Show that the base of the triangle is 12 cm.

| | | |
|---|---|---|
| **Step 1** | Area of parallelogram: base × height | $9 \times 4 = 36$ |
| **Step 2** | Triangle: half × base × height | $\frac{1}{2} \times b \times 6 = 36$ |
| **Step 3** | Simplify | $3b = 36$ |
| **Step 4** | Divide by 3 | $b = 12$ |
| **Step 5** | Answer | $\text{the base is } 12 \text{ cm}$ |

## Practice: a parallelogram and a triangle with the same area

**9.1** A parallelogram has base 12 cm and height 5 cm. A triangle has the same area and a height of 8 cm. Show that the base of the triangle is 15 cm.

**9.2** A parallelogram has base 15 cm and height 4 cm. A triangle has the same area and a height of 10 cm. Show that the base of the triangle is 12 cm.

**9.3** A parallelogram has base 7 cm and height 6 cm. A triangle has the same area and a height of 12 cm. Show that the base of the triangle is 7 cm.

**9.4** A parallelogram has base 10 cm and height 9 cm. A triangle has the same area and a height of 15 cm. Show that the base of the triangle is 12 cm.

**9.5** A parallelogram has base 16 cm and height 3 cm. A triangle has the same area and a height of 6 cm. Show that the base of the triangle is 16 cm.

**9.6** A parallelogram has base 9 cm and height 8 cm. A triangle has the same area and a height of 16 cm. Show that the base of the triangle is 9 cm.